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DETAIL SPECIFICATION

# LEATHER, CATTLEHIDE, FOR GLOVES

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

## 1. SCOPE

1.1 Scope. This specification covers the requirements for full grain cattlehide leather for gloves.

1.2 Classification. The leather is available in the following types, classes and treatment as specified (see 6.2).

1.2.1 Types.

Type I - Light, 2.0 to 3.0 ounces
Type II - Heavy, 3.0 to 4.0 ounces

1.2.2 Classes.

Class A - Sides
Class B - Bends

1.2.3 Treatment.

Treatment A - Water resistant treated

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

AMSC N/A FSC 8330

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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

(Copies of these document are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1517 - Standard Terminology Relating to Leather
ASTM D1813 - Standard Test Method for Measuring Thickness of Leather Test Specimens
ASTM D2211 - Standard Test Method for Elongation of Leather
ASTM D2617 - Standard Test Method for Total Ash in Leather
ASTM D2807 - Standard Test Method for Chromic Oxide in Leather (Perchloric Acid Oxidation)
ASTM D2810 - Standard Test Method for pH of Leather
ASTM D2813 - Standard Practice for Sampling Leather for Physical and Chemical Tests



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ASTM D2821 - Standard Test Method for Measuring the Relative Stiffness of Leather by Means of a Torsional Wire Apparatus
ASTM D3495 - Standard Test Method for Hexane Extraction of Leather
ASTM D4705 - Standard Test Method for Stitch Tear Strength, Double Hole
ASTM D5052 - Standard Test Method for Permeability of Leather to Water Vapor
ASTM D5053 - Standard Test Method for Colorfastness of Crocking of Leather
ASTM D6012 - Standard Test Method for Determination of Resistance of Leather to (Bleeding) Color Stain Transfer
ASTM D6014 - Standard Test Method for Determination of Dynamic Water Absorption of Leather Surfaces
ASTM D6076 - Standard Test Method for Shrinkage Temperature of Leather

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. The leather, when embossed, shall match the standard sample for fine cell pattern and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3), when tested as specified in 4.6.

3.3 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material. The Types I and II leather shall be full grain, chrome tanned cattlehide sides. The Type I leather shall be produced from thin, spready hides selected from native dairy cows. The type II leather shall be produced from any suitable cattlehides, except bull hides.

3.5 Trim (form) and area. The Class A leather shall be trimmed to conform with the definition of a “side” and the Class B leather shall be trimmed to conform to the definition of a “bend” as specified in ASTM D1517. The area expressed in square feet shall be legibly marked in the butt area on the flesh side of the leather.



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3.6 Color. The color of the Type I leather shall be black. Unless as specified in (see 6.2), the color of the Type II shall be cream, and shall be struck through from grain to flesh. The colored leather shall meet the colorfastness requirements as specified in Table I when tested as specified in 4.5.

3.7 Finish. The leather shall be soft and pliable. Application of a finish to the grain surface shall be prohibited. If necessary, the flesh side of the leather shall be dry buffed and shaved to obtain a smooth, clean surface and uniform thickness in all areas of the side or bend. Variations in color on the flesh side resulting from buffing or shaving are permissible. A fine hair cell embossed pattern may be applied to the grain surface of the side. Printed leather shall not be permitted. If embossed, the leather shall match the standard for fine hair cell pattern. Leather previously dyed or finished in a color not specified and then re-dyed or refinished to match a specified color shall not be acceptable.

3.8 Physical requirements. The finished leather shall conform to the physical and chemical requirements listed below in Table I, when tested as specified in 4.5.

TABLE I. Physical and chemical requirements.

| Characteristic | Requirement | |
|---|---|---|
| | Type I | Type II |
| Thickness (ounces) 1/<br>Min.<br>Max. | <br>2.0<br>3.0 | <br>3.0<br>4.0 |
| Colorfastness to crocking (min.):<br>Dry<br>Wet | <br>4.0<br>3.5 | <br>4.0<br>4.5 |
| Shrinkage temperature (≥ 98 ± 0.5 Celsius) @ 30 seconds 2/ | Pass/Fail | Pass/Fail |
| Stitch tearing strength, (lbs) (min.) 3/ | 17 | 30 |
| Elongation (percent) at 25 pounds (min.)3/, 4/ | 25 | 25 |
| Stiffness, (degrees), (max.) 3/<br>Before soaking<br>After soaking | <br>120<br>135 | <br>260<br>320 |
| Hexane-soluble material, (percent) (max.) | 25 | 25 |
| Total ash, (percent) (max.) | 9.0 | 9.0 |
| Chromic oxide, (percent) (min.) | 3.0 | 3.0 |
| pH (min.) | 3.3 | 3.3 |

1/ 1 ounce = 1/64 inch.
2/ Shrinkage shall not occur before it reaches that temperature or when held at that temperature for 30 seconds.
3/ At least 80 percent of the specimens shall meet this requirement.
4/ Any specimen which ruptures or exhibits grain crack below 25 pounds shall be reported as a failing specimen.



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3.9 Treatment A. When Treatment A is specified (see 6.2), the leather shall be treated with a water resistant compound. After the water resistant compound has been applied, the leather shall not be exposed to soaps, detergents, wetting agents, or surfactants.

3.9.1 Staining. When Treatment A is specified (see 6.2), the leather shall be tested for staining in accordance with 4.5. The stain on the test pad shall not be lower than a rating of 3.5 when tested as specified in 4.5.

3.9.2 Water absorption resistance. When Treatment A is specified (see 6.2), at least 80 percent of the specimens tested shall gain not more than 20 percent of their weight by water absorption when tested as specified in 4.5.

3.9.3 Moisture vapor transmission rate (MVTR). When Treatment A is specified (see 6.2), the leather shall meet a minimum MVTR of 500 $g/m^2$/24 hours when tested as specified in 4.5.

3.10 Workmanship. The finished leather shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract or purchase order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination of the end item. The leather for each type shall be examined and defects shall be classified as stated in Table II. The sample unit for these examinations shall be one (1) hide, side, skin, front or bend as required for each type specified. The contract or purchase order shall identify the number of samples required for thickness for acceptance and rejection of the leather skins. Separate examinations shall be performed for visual and the thickness, but the same sample may be used for each examination.



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TABLE II. Visual examination defects.

| Examination | Defect 1/ |
|---|---|
| Trim<br>Class A<br>Class B | <br>Not sides<br>Not bends |
| Color (all types) | Not as specified<br>Color not uniform on grain side<br>Color of Type I leather does not completely penetrate leather from grain side through flesh side |
| Finish | Not full grain<br>Flesh side not smooth or contains areas of coarse loose fiber<br>Grain surface has application of finish<br>Embossing other than a fine hair cell pattern Printed design on leather<br>Does not match standard sample for fine cell pattern |
| Quality of leather | Not clean; stain or other foreign matter.<br>Hard, boney, loose, spongy leather, hard scar, cut, hole, brand, scratch, deep, fat wrinkle, or grain damage which causes more than 12 percent loss in total area of a side or more than a six (6) percent loss in total area of a bend |
| Marking | Omitted, illegible, or not in area specified |
| Area expressed in terms of square feet | Less than area marked on the side or bend |

1/ Light, well healed scratches and grub holes, light fat wrinkles, and slight stains shall not be classified as defects.

4.5 End item testing. The finished leather shall be tested for the characteristics as specified in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. The sampling location from which the sample unit is to be obtained shall be in accordance with ASTM D2813. The area of the unit of product from which the specimens are taken shall be 12 by 12 inches. Fifteen units of product shall be selected at random from each lot for the purpose of testing. For lots consisting of less than 15 units of product, each unit shall be sampled for testing. The sample(s) required for testing the lot for physical and chemical characteristics shall be selected from the samples for visual or dimensional thickness examination. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any of the following conditions exists:

a. More than three (3) failures occur for elongation.
b. More than three (3) specimens fail to meet stitch tearing strength requirements.
c. More than three (3) specimens fail to meet stiffness requirements.
d. More than three (3) specimens fail to meet water absorption requirements.
e. One or more specimens fail to meet any of the remaining requirements applicable to the sample unit.
f. Any composite fails to meet the specified requirement.



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TABLE III. End item tests.

| Characteristic | Requirement Reference | Test Method | Requirement | |
|---|---|---|---|---|
| | | | Sample unit | Composite sample |
| Material | See 3.4 | Visual (see 6.4) | - | - |
| Tannage | See 3.4 | Visual (see 6.4) | - | - |
| Color | See 3.6 | See 4.6.1 | - | - |
| Finish | See 3.7 | Visual | - | - |
| Thickness | Table I | ASTM D1813 | X | - |
| Staining | Table I & 3.9.1 | ASTM D6012 1/ | X | - |
| Colorfastness to crocking | Table I | ASTM D5053 1/ | X | - |
| Elongation | Table I | ASTM D2211 | X | - |
| Stitch tear strength | Table I | ASTM D4705 | X | - |
| Shrinkage temperature | Table I | ASTM D6076 | X | - |
| Stiffness | Table I | ASTM D2821 & 4.6.2 | X | - |
| Hexane-soluble material 2/ | Table I | ASTM D3495 | - | X |
| Total ash 2/ | Table I | ASTM D2617 | - | X |
| Chromic oxide 2/ | Table I | ASTM D2807 | - | X |
| pH | Table I | ASTM D2810 | - | X |
| Water absorption resistance (Treatment A) | See 3.9.2 | ASTM D6014 | X | - |
| Moisture vapor transmission rate (Treatment A) | See 3.9.3 | ASTM D5052 | X | - |

1/ Evaluated using AATCC EP8.
2/ Calculated on moisture-free basis.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the finished leather shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with sources simulating artificial daylight $D_{75}$ illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

4.6.2 Stiffness after soaking. Any individual specimen exceeding the maximum stiffness values specified before soaking shall not be tested for stiffness after soaking. The specimens shall be immersed in 1000 milliliters of distilled water at 140 ºF (± 5)ºF (60 ºC (± 3)ºC for 20 (± 2) hours. Complete immersion of the specimens shall be accomplished by placing a floating cover on the vessel used for immersion of the specimens. After the required immersion period, the specimens shall be removed from the water and reconditioned in accordance with standard atmospheric conditions as specified in the test method. The reconditioned specimens shall be placed flesh side down on a ruler graduated in decimal inches and the distance between the benchmarks shall



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be measured to the nearest 0.05 inch. If the distance is 3.4 to 3.6 inches, test the specimen in accordance with ASTM D2821. If the distance is 3.2 inches or less, the specimen shall be reported as a test failure. If the distance between benchmarks is 3.25 to 3.35 inches, the specimen shall be stretched before testing. To stretch the specimen, place the specimen on the ruler and align line one of the benchmarks with a division line on the rule. Hold the specimen against the ruler by placing the fingertip of one hand on the aligned benchmark. With the thumb and index finger of the other hand, lightly grip the specimen near the aligned benchmark, then slide the finger and thumb along the specimen to the other bench mark while exerting pressure on the specimen that is sufficient to obtain the required distance between the bench marks. If the required distance between the benchmarks has been obtained (3.4 to 3.6 inches), test the specimens in accordance with ASTM D2821. If the distance between the benchmarks is greater than 3.6 inches discard the specimen and test a new specimen selected from the same 8 by 8-inch test sample of leather from which the discarded specimen was obtained. The new specimen shall be tested for stiffness before and after soaking. The test results obtained when the discarded specimen was tested for stiffness before soaking shall not be reported.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I leather is intended for use in the fabrication of light duty gloves and Type II leather is used in the fabrication of heavy duty gloves.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class and if treatment is required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1).
e. Color of leather required (see 3.6).
f. Conformance inspection (see 4.3).
g. Inspection conditions (see 4.3.1).
h. Packaging (see 5.1).



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6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

6.6 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.7 Subject term (key word) listing.

Bends
Chrome tan
Full grain
Sides
Water resistant

Custodians:
Army - GL
Navy - NU
Air Force 11

Preparing Activity:
DLA-CT

(Project: 8330-2021-001)

Review activities:
Army - MD
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.



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